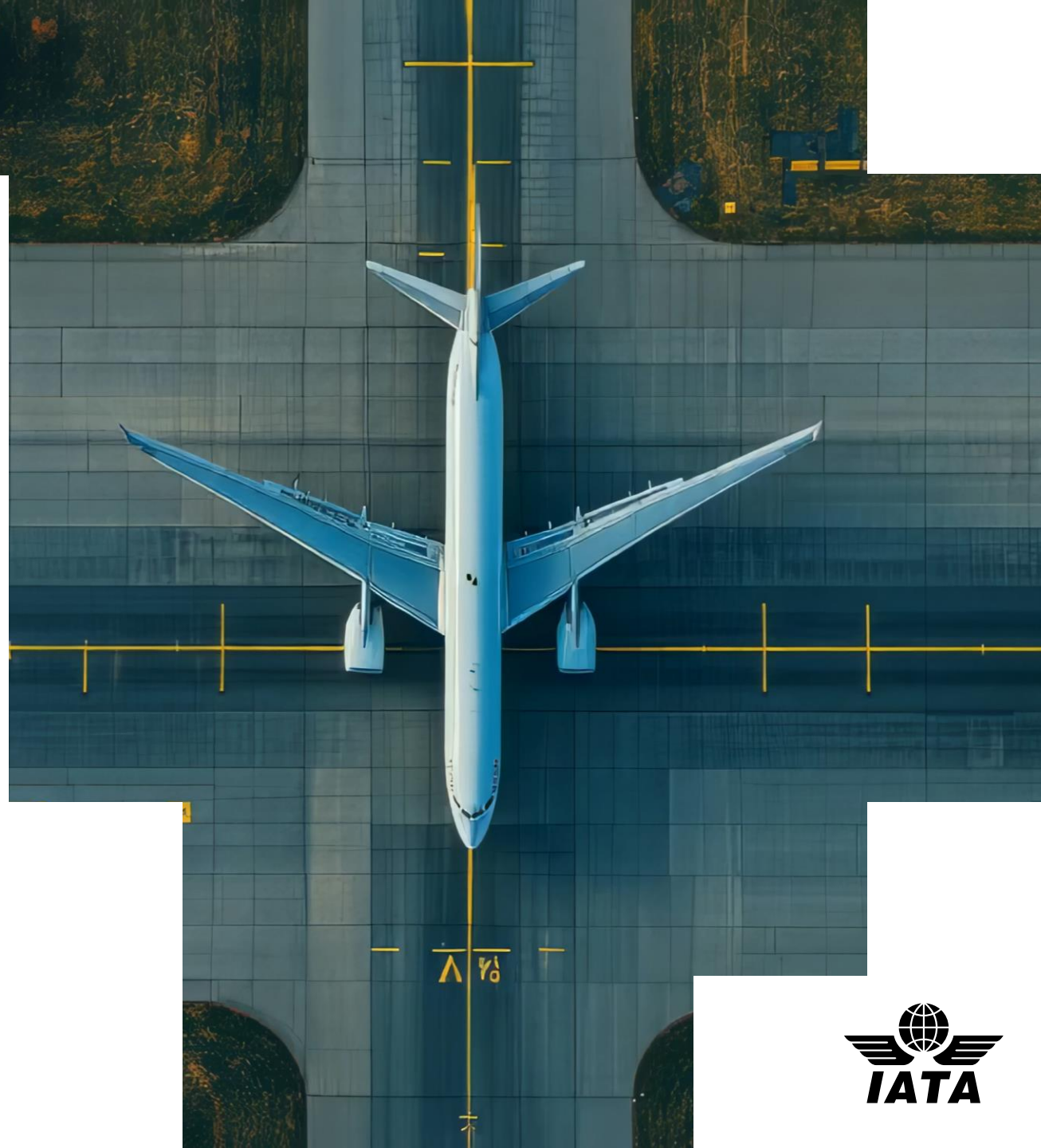
Global Aviation Security in the swiyu Trust Infrastructure

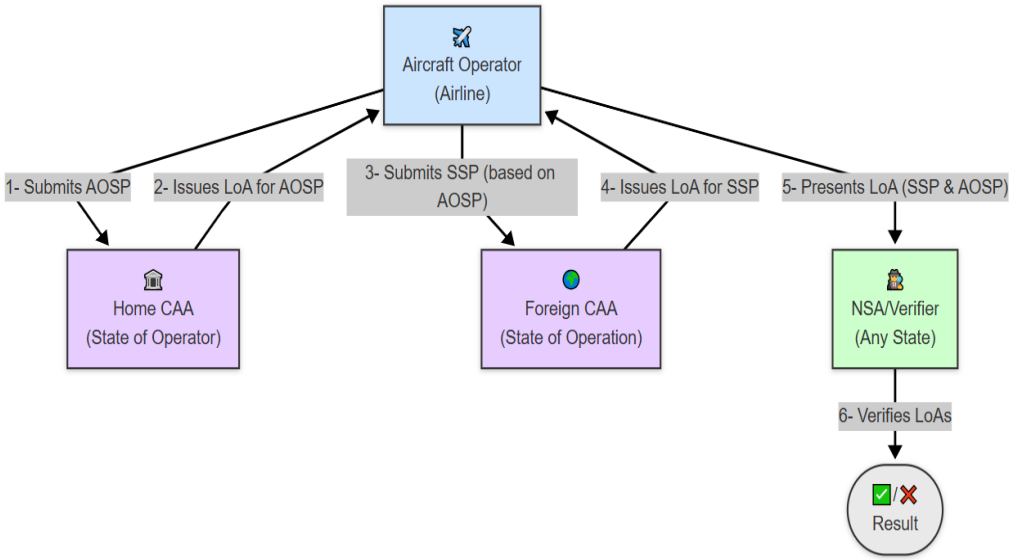
The IATA ASTF Initiative

Réunion participative swiyu

27 Feb 2026



Why Aviation Security Trust Framework Matters



ICAO sets the global AOSP and SSP standards; Member States mandate and enforce them

- Current aviation security approvals, Aircraft Operator Security Programs (AOSP) and Supplementary Station Procedures (SSP), rely on manual, document-heavy processes
- Lack of cross-border interoperability, and deficiencies in digital trust between Civil Aviation Authorities (CAAs) and foreign National Security Authorities (NSAs).
- To transition aviation to a digital trust model, ensuring AOSP and SSP approvals are genuine and instantly verifiable across borders without compromising autonomy.

Build on decentralized Open standards

Standardization & Interoperability

Establishing clear, unified digital standards for attestation, allowing verifiable approvals to be instantly verified and trusted across organizational and national jurisdictions.

swiyu

Neutral Trust Infrastructure

The framework is anchored in the neutral, state-operated Swiss Public Trust Infrastructure (swiyu).

Digital Autonomy

Offering stakeholders a path toward a coordinated digital future without compromising the autonomy and control of their existing systems and processes.

IATA Aviation Security Trust Framework

Verifiable Credentials (VCs)

Think of a VC as a tamper-proof digital seal of approval. We use it to turn a document's status (like CAA Approval) into portable, machine-readable proof that can instantly be checked for authenticity.

DID

Decentralized Identifiers (DIDs)

We utilize Decentralized Identifiers (DIDs) as a digital anchor for CAAs, allowing cross-border trust without relying on a single central certificate authority.

Cryptographic verifiability and binding

Guaranteeing that approvals are genuine, unaltered, and verifiably linked to the specific document version they approve.

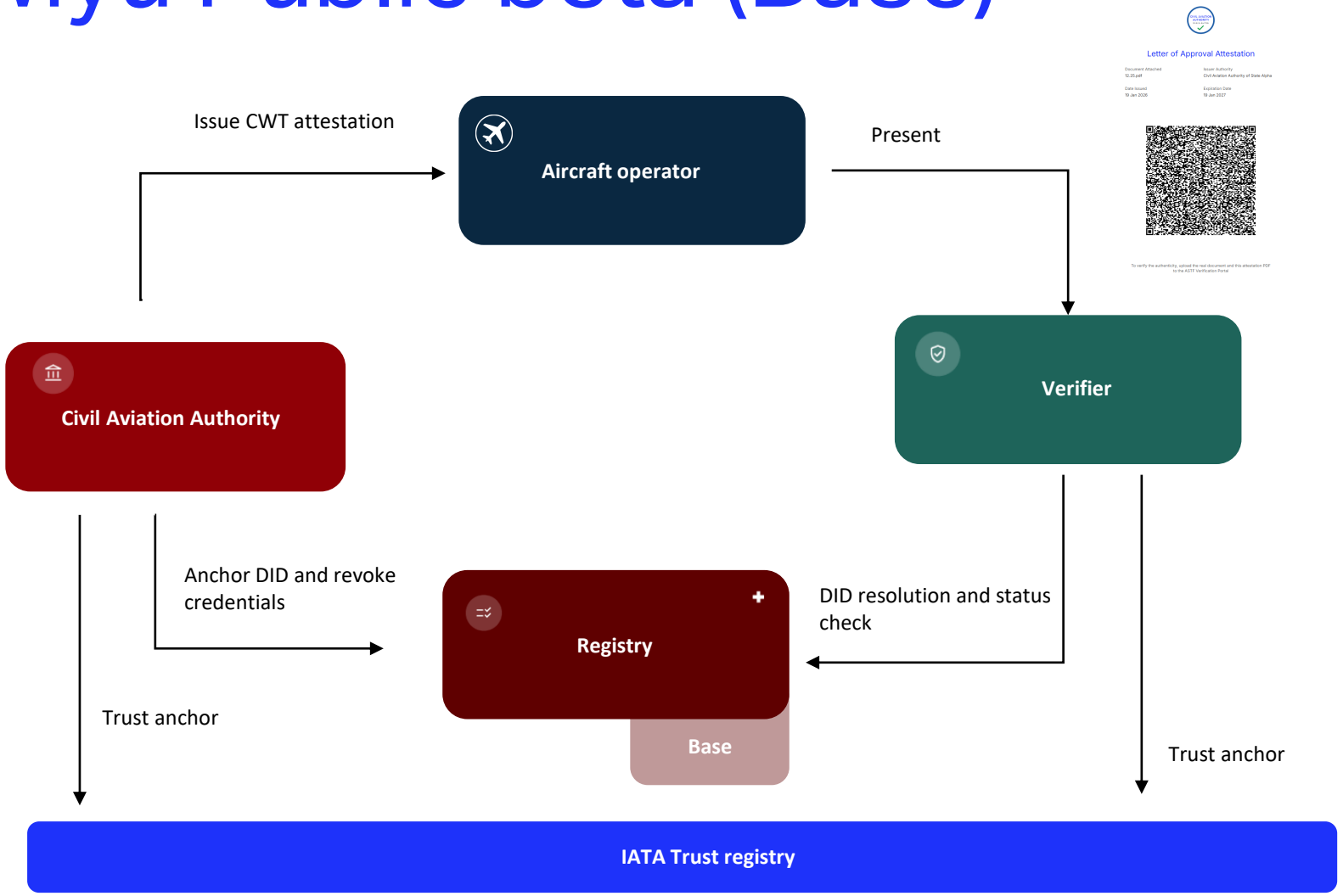
Anchoring in Swiyu Public beta (Base)

Anchor Public Keys

Securely register the cryptographic public keys
We utilize the **did:tdw** (Trust DID Web v0.3) method to securely register the globally unique identifiers of Civil Aviation Authorities (CAAs) and associated cryptographic public keys .

Status management

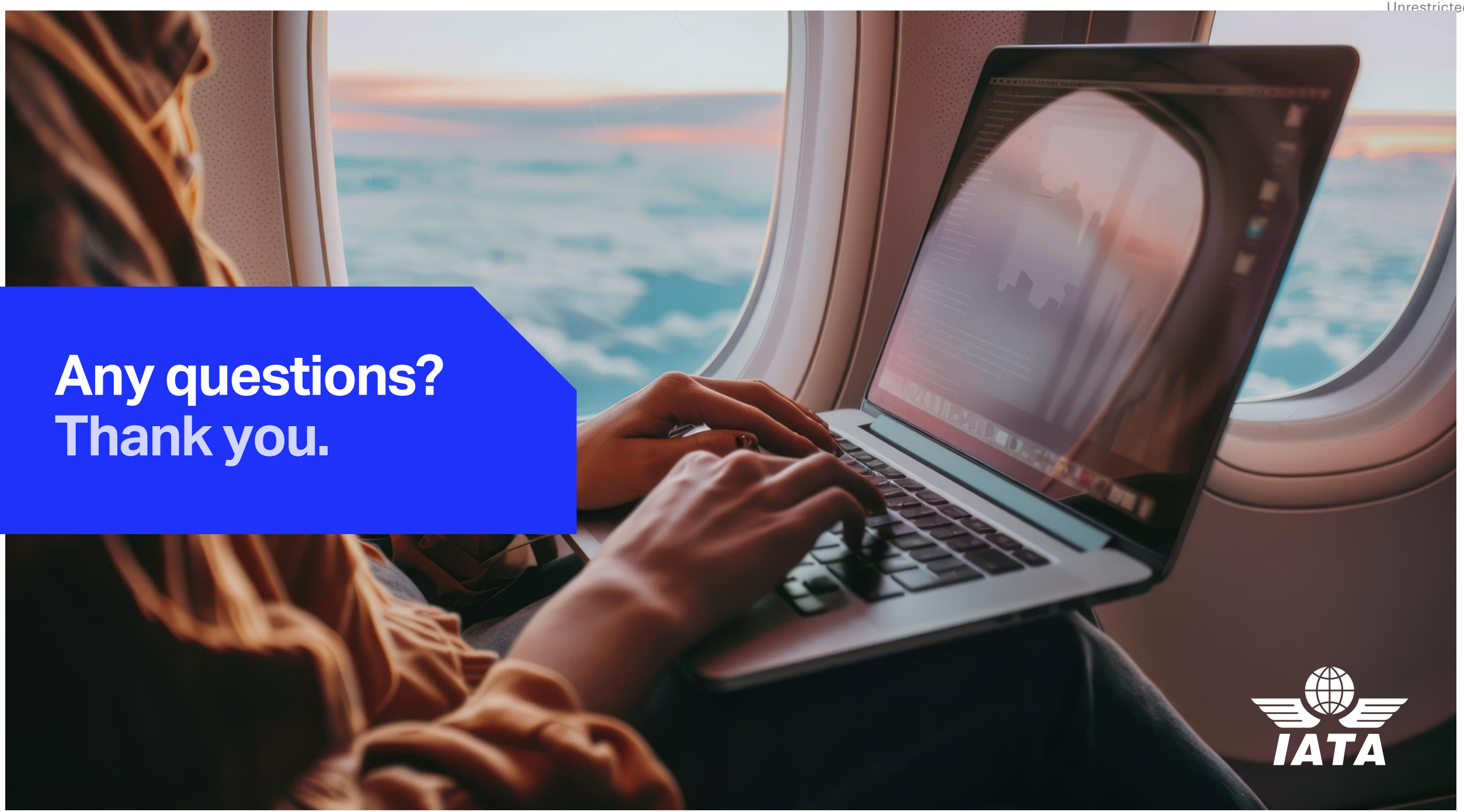
We leveraged Bitmask-based Token Status Lists to enable verifiers to perform real-time revocation checks of any document.



The ASTF Demonstrator in action

Feature	Implementation
Attestation-Centric Credentials	The ASTF Proof of Concept issues "**Letters of Approval**" as machine-readable Verifiable Credentials.
Compact QR Codes	These approvals are compressed using the CBOR Web Token (CWT) format, allowing them to be embedded directly into scannable QR codes for both digital and physical substrates.
Real-Time Revocation	We manage the lifecycle and real-time validity of these credentials using Bitmask-based Token Status Lists.
Instant Verification	Any verifier (like an NSA inspector) can instantly resolve the CAA's public key via the swiyu infrastructure to validate the digital signature, the document hash, and the active status of the token.

Unrestricted



Any questions?
Thank you.